

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	4

Case Study Topic: Delhi-NCR Delivery Routing using Bidirectional Dijkstra Algorithm

Word Done Summary:

In this case study, we analyzed route optimization in the Delhi-NCR delivery network using Bidirectional Dijkstra Algorithm. The road network was represented as a weighted graph where intersections acted as vertices and travel times acted as edge weights. We compared standard Dijkstra's algorithm with bidirectional Dijkstra to observe improvements in shortest path computation. The study also examined search expansion, meeting points, time complexity, and correctness conditions for bidirectional search.

Implementation Details:

1. Represented Delhi-NCR road network as a weighted graph.
2. Applied standard Dijkstra's shortest path algorithm.
3. Implemented bidirectional Dijkstra using forward and backward searches.
4. Maintained separate distance maps for both directions.
5. Identified the optimal meeting point between searches.
6. Compared efficiency between standard and bidirectional Dijkstra.
7. Studied correctness and stopping conditions for bidirectional search.

Result:

Bidirectional Dijkstra successfully reduced graph exploration while maintaining the same shortest path result as standard Dijkstra. The algorithm demonstrated faster practical routing performance for road-network applications.

Output:

```
Shortest distances from CP:  
Node 0 : 0  
Node 1 : 15  
Node 2 : 12  
Node 3 : 18  
Node 4 : 37  
Node 5 : 28  
Node 6 : 32  
Node 7 : 41
```

GitHub Link:

<https://github.com/ketanasindhusomisetty/DSA-Case-Studies/tree/main>

Student Signature	Faculty Signature
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